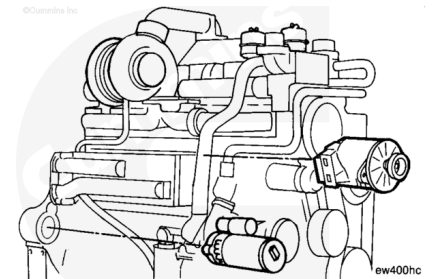


Trainings ▾

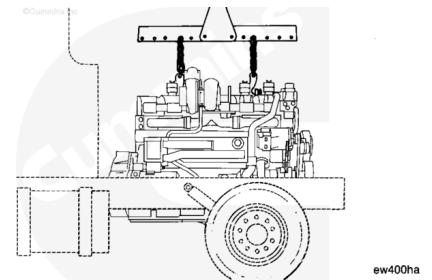
Install

Install all accessories and brackets that were removed from the engine.



⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Inspect the engine lifting brackets for damage or cracks. Do **not** attempt to lift engine if cracks or damage is visible.

Use engine lifting fixture, Part Number 3163264, or equivalent, for installation of engine. The fixture **must** be designed to lift a maximum 3175 kg [7000 lb]. Refer to manufacturer's instructions. Refer to Section V for wet and dry weight of engine.

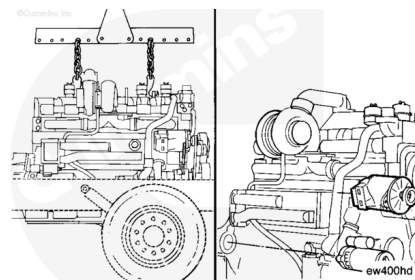
Use a properly rated hoist and engine lifting fixture to support the weight of the engine. Install the front and rear engine support capscrews.

Use a properly rated hoist, engine lifting fixture, Part Number 3163264, or equivalent, to install the engine into the chassis.

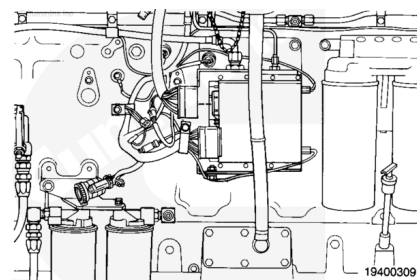
Align the engine in the chassis and tighten the engine mounting capscrews. Refer to the equipment manufacturer's torque specifications.

Connect all engine and chassis mounted accessories that were removed.

Make sure all lines, hoses, and tubes are in good condition, properly routed, and fastened to prevent damage.



Connect the OEM wiring harness to the electronic control module (ECM). Refer to Procedure 019-072 in the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html>).



⚠ CAUTION ⚠

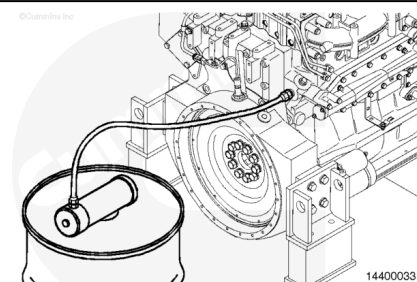
The lubricating system must be primed before operating the engine after rebuild to reduce the possibility of internal engine damage. Do not prime the system from the bypass filter as the filter will be damaged.

Remove the large plug [1-1/4 inch pipe thread] from the oil cooler housing.

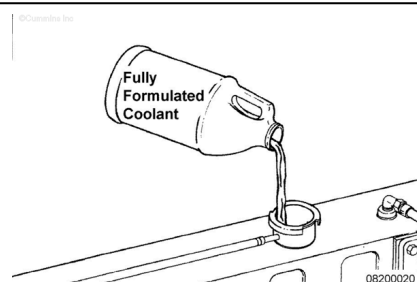
Use a pump capable of supplying 207 kPa [30 psi] continuous pressure. Connect the pump to the block at the rear of the engine oil cooler as shown.

Use a supply of clean 15W-40 engine oil. Turn the pump to the ON position.

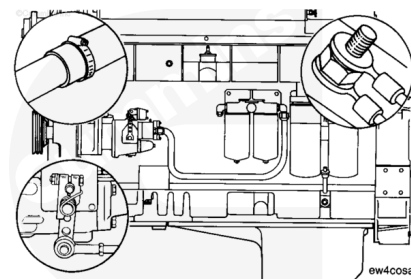
Check the engine oil pressure gauge. When the gauge indicates oil pressure, begin monitoring the oil level in the oil pan.



Fill the cooling system with fully formulated coolant. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/89/89-008-018.html>) and the Operation and Maintenance Manual, QSK23 Engine, Bulletin 4021374 (</qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021374.html>) for coolant specifications.

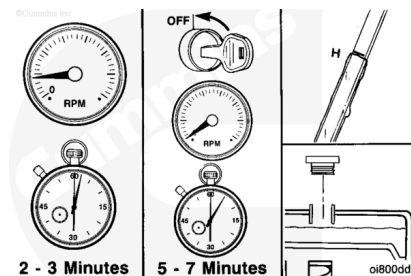


Complete a final inspection to be sure that all hoses, wires, linkages, and components have been properly installed and tightened.



⚠ WARNING ⚠

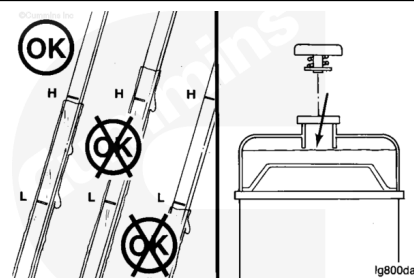
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine at low idle for 2 to 3 minutes.

Turn the keyswitch to the OFF position and wait 5 to 7 minutes for the oil to drain to the oil pan and check the oil and coolant levels again.

Add oil or coolant to obtain the correct level if necessary. Refer to Section V for oil specifications.



Operate the engine for 8 to 10 minutes to check for proper operation, unusual noises, coolant, fuel, and lubricating oil leaks.

Repair all leaks and component problems. Refer to the appropriate procedures.

